

Samkit Bothra

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EDUCATION

Florida Atlantic University, Boca Raton, FL

August 2023 - Present

Dual-enrolled Bachelor of Science in Computer Science

Minor in Business

- 4.0 /4.0 GPA

Florida Atlantic University High School, Boca Raton, FL

May 2026

Florida Atlantic University High School is an extremely competitive and rigorous program where students are fully dual-enrolled at Florida Atlantic University from tenth grade through twelfth grade; students take college courses to fulfill high school requirements. Most students graduate from FAU High School with 80-90 college credits.

- 3.9/4.0 GPA

LEADERSHIP

Student Mayor, Mayor, Parkland Florida

September 2024 – Present

- Serving as the official student representative of the city, advocating for youth perspectives and collaborating with city leaders on initiatives that impact local communities.
- Planning, hosting, and participating in citywide events, fostering community engagement, and promoting civic involvement among residents.
- Developing public speaking, project management, and leadership skills through active involvement in city government activities and public forums.

Jain Center of South Florida (IRCC) Youth Group, President

December 2024 - Present

- Plan and oversee the execution of large-scale celebrations for major Indian holidays, attracting over 500 attendees.
- Design and deliver interactive lessons on Hindu traditions and values, fostering cultural awareness, pride, and community engagement among children.
- Developing strong leadership, project management, and communication skills

INSTRUCTIONAL EXPERIENCE AND MENTORING

College Mathematics Tutor, Math Learning Center, FAU, FL

January 2025 – May 2025

- Providing one-on-one and small-group tutoring for college-level algebra, enhancing students' foundational understanding and problem-solving abilities.
- Tailoring tutoring sessions to accommodate diverse learning styles, integrating real-life applications and interactive exercises that foster engagement and mastery of complex concepts.
- Guided students in developing study plans, time management skills, and test-taking strategies.

FAU High Peer Mentor, FAU, Boca Raton, FL

August 2024 – May 2025

- Mentored high school underclassmen in navigating Florida Atlantic University's dual enrollment process, offering personalized guidance on registration, course selection, and university protocols.
- Provided ongoing consultation and support to address academic difficulties, including resolving issues with professors and managing class requirements.
- Developed and refined key skills in communication, problem-solving, and mentoring, ensuring students receive the necessary tools and confidence to succeed.

STAR Lab Instructor, FAU High School, *Boca Raton, Florida*

January 2023 -Present

- Instructing young researchers and assisting them in their projects
- Taught younger researchers about basic lab skills and etiquette

Kumon, Coral Springs, FL

August 2023 – January 2024

- Helped young kids to improve their math and reading skills (be more specific)
- Gained instructional experience in teaching and instructing

CONFERENCES AND SYMPOSIUMS

BristleCon 2025, San Francisco, *California*

January 22 - January 24, 2025

- Engaged with industry-leading experts from Sofar Ocean, Bristlemouth, and other prominent organizations in ocean exploration and technology.
- Presented research project on Underwater Acoustic Sensing and Wireless Communications to CEOs and startup founders, aiming to secure strategic partnerships for further innovation.
- Collaborated with senior engineers at Sofar Ocean and Bristlemouth, receiving valuable feedback and brainstorming enhancements to advance the project's technological potential.

Florida Undergraduate Research Conference, USF, Tampa, *Florida*

February 14 – February 15, 2025

- Presented “Exploring Gender Differences in Perceptions of AI’s Impact on Social Media” at the Florida Undergraduate Research Conference (FURC), a study examining how men and women differ in their views on misinformation, privacy, and the societal influence of AI-driven content.
- Highlighted the growing importance of understanding demographic perceptions in the age of AI to inform ethical tech development and responsible social media design.
- Connected with experts in the field and received vital insights on how to further develop and expand the project.

FAU OURI Symposium, FAU, Boca Raton, *Florida*

April 4, 2025

- Gave a 12-minute oral presentation to judges on my AI’s influence on misinformation in social media polling project.
- Awarded a first-place certificate along with a \$100 cash prize for the importance and presentation style of the project.
- Also held a poster presentation on my biomechanics project titled “Validation of a Wearable IMU Sensor against the Optotrak Motion Capture Device for Assessment of Head and Neck Tracking”

National Conference on Undergraduate Research, Pittsburgh, *Pennsylvania*

April 7 – April 9, 2025

- Delivered a formal oral presentation on my research analyzing gender-based differences in public perceptions of AI’s influence on social media, misinformation, and privacy
- Focused on the methodology and statistical findings of a 16-question survey distributed to over 200 respondents
- Connected with experts from all around the nation in various fields and received vital insights on how to further develop and expand the project.

PROFESSIONAL DEVELOPMENT

Young Scholars Program, *Member*, Florida State University, Tallahassee, *Florida*

June 7 – July 19, 2025

- One of the 40 students selected into this six week STEM summer program out of a pool of hundreds of students across Florida.
- Took classes such as Quantum Computing and Game Theory
- Worked with an FSU professor as a Molecular Biophysics Researcher

National Honor Society, Member, FAU High School, Boca Raton, Florida

October 2023-Present

- Went to several beach cleanups to raise awareness of pollution.
- Learned about college scholarships and what is required to receive them.
- Helped others to apply for and research which scholarships they would want to apply to.

Key Club, Secretary, FAU High School, Boca Raton, Florida

September 2023 - Present

- Organizing volunteer events such as food for the poor.
- Presenting to members of the club and getting them to participate in volunteer activities.

ACADEMIC PROJECTS

MD Simulations to Uncover Cryptic Sites in SARS-CoV-2 Protease, YSP, FSU

June 7 – July 19, 2025

- Performed molecular dynamics simulations of the SARS-CoV-2 main protease to identify cryptic allosteric pockets not visible in static structures.
- Using enhanced sampling techniques, we ran a 300 ns simulation and successfully identified a transient pocket at the dimer interface.
- This hidden pocket could serve as a novel drug target for future COVID-19 antiviral development, offering a new strategy to inhibit viral replication.
- Tools used:
 - Linux
 - Visual Molecular Dynamics
 - CHARMM-GUI

Underwater Acoustic Sensing and Wireless Communications, CAAI, FAU

June 2023 - Present

- Conducting a cutting-edge research project on Underwater Acoustic Sensing and Wireless Communications, aimed at enhancing ocean surveying methods and data accuracy.
- Delving into technical innovation by integrating advanced sensor arrays, signal processing algorithms, and robust software solutions to enable reliable, real-time underwater data transmission.
- Presented key findings at a high-profile conference, engaging directly with CEOs and leading experts in ocean exploration and technology.
- Tools used:
 - WinSCP
 - Python
 - C Programming
 - Linux
 - Raspberry PI
 - Terminal Command Prompt

Wearable Sensors for Tracking Body Motion: Advancing Rehabilitation, FAU

January 2025 – Present

- Developed and tested a wearable IMU-based system to track head and neck movement for assessing fall risk and postural control in patients
- Used AI-driven analysis and 3D modeling to reconstruct movement patterns and visualize deviations in the center of mass over time
- Validated sensor data by comparing it against the gold-standard Optotrak motion capture system using vector angle comparisons.
- Tools used:
 - Python
 - Lenovo 3D-camera software
 - Excel
 - Optotrak
 - First Principles

Business Economic Polling Initiative, Florida Atlantic University

August 2024 – Present

- Conducted an in-depth survey-based research study on “Exploring Gender Differences in Perceptions of AI’s Impact on Social Media,” aiming to uncover how diverse genders perceive AI’s growing societal relevance.
- Collected and analyzed data from over 200 respondents, refining statistical analysis and data visualization skills to draw meaningful insights on user attitudes and experiences.
- Tools Used:
 - Statistical Package for the Social Sciences (SPSS)

Summer Lab Skills Workshop, Pompano Beach, FL 33062

June 19-23, 2023

- Conducted field surveys by gathering microplastics from Wahoo Bay.
- Gained skills in teamwork through working with a team of college graduates to collect data.
- Gained skills in leadership through leading a team through sample analysis.

RELEVANT COURSEWORK

Research Methods, Business Economic Polling Initiative (BEPI), Data Structures & Algorithms, Computer Logic Design, Physics for Engineers, Calculus 1 & 2, Accounting

SKILLS**Programming Languages**

- Python, C++, C

Technical Skills

- Linux
- Raspberry PI software integration
- Cellular data transmission
- Data Science Fundamentals
- Underwater Robotics

Languages

- **English** (100% Proficiency), **Hindi** (100% Proficiency), **Spanish** (50% Proficiency)

HONORS AND AWARDS**Distributive Education Clubs of America International Qualifier**

- Placed third in the entire state in March 2025 for exam + roleplay performance and qualified for the International Career Development Conference (DECA ICDC).
- Qualified for states in January 2025 with a first-place award in district event for the season, with a 90 on the exam and a 98 on the roleplay, and recognition from the school for the highest average in the chapter.
- Excelled in role-play acting as a financial advisor and gaining skills in communication and business

Several Debate Tournament Wins

- Participated and placed in several district wide debate tournaments in Lincoln Douglas.
- Gained many communication skills through speaking in front of big crowds.

Two-time State Golf Qualifier and One-time State Champion

- Playing on the high school varsity golf team at the number 3/4 spot on the five-team roster.
- Played several individual tournaments and placed 1st twice and 2nd three times in South Florida .

